

Analysis of the Seine-Normandy river basin, using WFD criteria



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1. Introduction

The Seine is a river in the North-West of France. The river originates on the Mont Tasselot, which lies 471 meters above sea-level in the Côte d'Or region of Burgundy. From its origins it runs from through Paris and empties at the city of Le Havre in the English Channel, covering roughly 80km. At Le Havre the Seine discharges 500 m³/s, which is fairly small compared to other rivers in the world. The Seine has several tributaries. The Oise, a right tributary which originates in Hainaut, Belgium, and reaches the Seine just west of Paris. The Marne is another right tributary that starts in the Langres Plateau. The Marne and the river Seine meet each other in Paris. And finally there is river Yonne. This is a left tributary which starts in the Mount Prenely. It meets the Seine in Montereay-Fault-Yonne which is south-east of Paris. The climate is temperate. With average annual rainfall 750 mm and temperatures running from an average of 2,5°C in winter to 24,6°C in the summer.



Figure 1.1: Seine-Normandy basin topography. Source: United Nations, 2003

The river is of high economic importance. A large part of the industry as well as agriculture is located in this basin, with the river providing a way of transport as well. Ocean going vessels can go up-river as far as Rouen, which is 120 km away from the sea. As a result the river has been under a lot of anthropogenic stress, resulting in changes to the river both hydromorphologically and biological. With the Water Framework Directive (WFD) in place, the condition of the river will need improve to achieve the goals set for 2015. In this paper the current status of the river, both chemical and ecological will be examined, as well as the threats, management and administrative organization, and future perspectives. The aim is to examine the river the way it is now and what it could possibly look like in the near future.

2. Hydromorphology

WFD definition

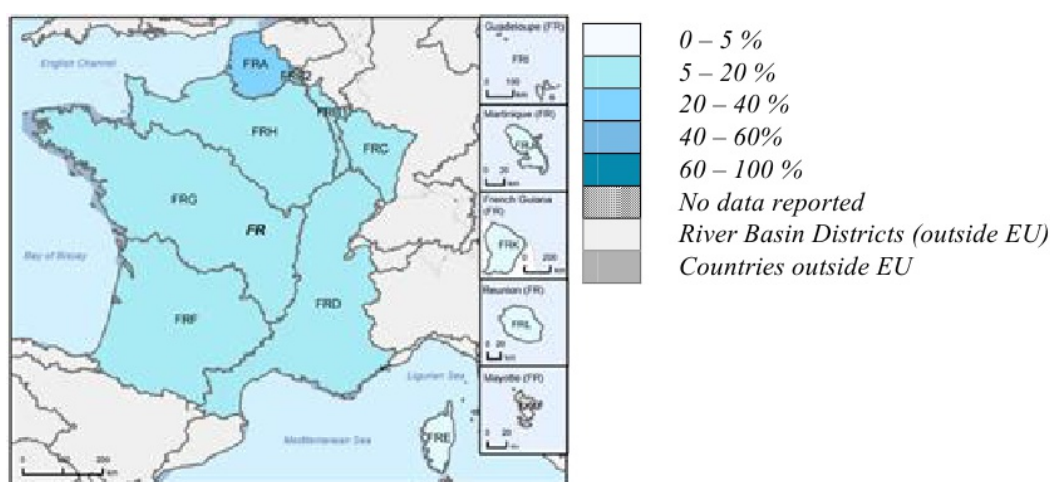
The Water Framework Directive defines hydromorphology as:

“Hydromorphology is used in river basin management to describe the hydrological and geomorphological processes and attributes of rivers, lakes, estuaries and coastal waters” (River Basin Management guide to hydromorphology no. 1, 2014).

Humans often alter the hydromorphology of a river to make it provide certain functions. For example dredging to allow larger vessels to move upstream, or artificial reservoirs for drinking water. However, changing the hydromorphology of a river can have serious impacts on the environment and the behavior of the river, often having an effect on the biology of the river. The WFD makes a distinction between two types of modified bodies of water, namely artificial water body (AWD) and heavily modified water body (HMWB). An AWD is a body of water that is entirely man-made. An example of this are canals. HMWB's on the other hand are already existing water bodies that, due to human influence, have been changed to fulfill a specific function. Some changes can even cause the water body to change in category. When a dam is built, part of the river can then become a reservoir (WFD and hydromorphological pressures. Technical report, 2006).

Good Ecological Potential

Another important thing to note is that when talking about a AWD or a HMWB, the GES will most likely never be achieved. AWD/HMWB's often have important socio-economic benefits that, if GES is to be achieved, would be lost. Because of this the WFD came up with a another term called good ecological potential (GEP). GEP is the ecological condition the water body can be in without losing its function (WFD and hydromorphological pressures. Technical report, 2006). GEP therefore is the main target for both AWD's and GMWB's. Because of the GEP the methods used to solve problems in AWD/HMWB's or often different than in other water bodies because the function of the water body has to be taken into account. For all of the french River Basin Districts (RBD's) the percentage of AWD/HMWB's has been identified. Figure 2.1 shows the overall percentage of AWD/HMWB's for all the RBD's in France.



The Seine-Normandie basin is indicated in the figure with the abbreviation FRH. As can be seen from this figure, 20-40% of water bodies in the Seine-Normandie basin are heavily modified or artificial (European Commission, 2012). Table 2.1 shows detailed data specifically for the basin.

RBD	Rivers		Lakes		Transitional		Coastal	
	No.	(%)	No.	(%)	No.	(%)	No.	(%)
HMWBs	48	2.86	16	35.6	6	85.71	2	10.53
AWBs	20	1.19	28	62.2		0		0

Table 2.1: Number and percentage of HMWB's and AWB's in France. Source: WISE

The designation of AWD/HMWB's is based on several uses of the water body. The main uses are power generation, navigation, flood protection, water regulation, irrigation, and recreational purposes (European Commission, 2012).

Hydromorphological issues

Over the years, the anthropogenic pressures have altered the Seine significantly and several problems have arisen. The runoff from the river has increased. This is due to the fact that part of the basin has become impermeable. As a result, less water penetrates into the ground, forcing water to move along the surface. The additional runoff is estimated at 760 million cubic meters per year (United Nations, 2003). In addition to this, the dams in the Seine-Normandy basin theoretically cannot cope with the volumes of extreme floods. These dams also cause problems to the biology in the river. Dams (and also weirs, locks, impoundments etc.) pose great obstacles for migratory fish and invertebrates that migrate up and down the river. Dams also impeded with the transport of materials such as organic and inorganic sediment. A solution for the migratory fish is the use of fish passes. However, less than 20 percent are in fact equipped with these passes. There is the same problem with hydroelectric power plants. 60% percent of the hydroelectric power plants are impassable for migratory fish (United Nations, 2003). Another important factor for the decline of the fish population is the geomorphological alterations of the rivers for navigation purposes. Fig 2.2 shows several consequences of such alterations.

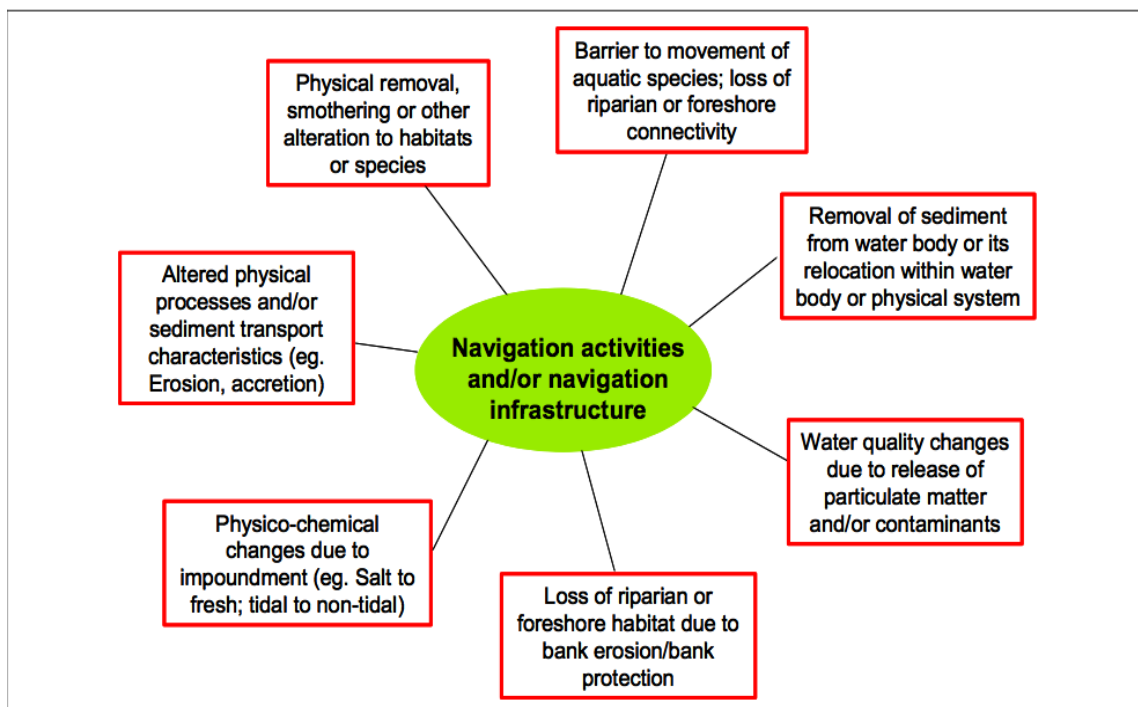


Figure 2.2: Consequences of water body modifications for navigation, Source: EC, 2006

There is still room for improving parts of the Seine-Normandy river basin when it comes to hydromorphology. Solutions to several problems already exist and in some cases, like the fish passes, have also been implemented. However, more work and investments are required to achieve the GEP of AWD/HMWB's.

3. Watertypes

Member states have defined surface water body types for each surface water category and have established type-specific reference conditions representing that type at high ecological status. The main purpose of this typology is to function as an anchor for the ecological status classification. Two systems are used (A and B), with B providing higher flexibility and ecological difference detection at higher implementation difficulty. System A provides a set of standard types and system B allows for more (sub)classifications based on chemical, physical, and other characteristics deemed important for regional ecological differentiation (Nixon et al, 2012).

System A water bodies are classified according to upstream catchment surface area (10-100, 100-1000, 1000-10000, >10000 km²), altitude (<200m, 200-800m, >800m) and geology (>50% calcareous geology - "calcareous", <10% calcareous geology - "siliceous", >50% histosols in upstream catchment area - "organic", or mixed). Histosols are highly organic soils such as peat and have a strong effect on organic content of the surface water.

Watertypes in France

France has reported 141 national water body types under System B classification. This is a high number compared to system A and other EU member states; France has also declared these have only been partly tested against biological data. Since the "Comparative Study of Pressures and Measures in the Major River Basin Management Plans" (Nixon, 2012) states: "It is important that the established typology system assists in achieving an adequate confidence in reference conditions and the subsequent classification of ecological status. In all cases the ecological relevance of the different theoretical types has to be demonstrated by cross-checking against biological data such as macroinvertebrate groups and/or species composition", this is unexpected.

Most of the reported types are river water bodies. Table 3.1 below reports the amount of water bodies France has reported for the most common types.

Altitude	Geology	Catchment area			
		Small	Medium	Large	Very large
Lowland	Siliceous	608	1146	117	6
	Mixed	66	184	59	2
	Calcareous	310	609	206	49
	Organic	1			
Mid-altitude	Siliceous	985	1364	303	70
	Mixed	108	277	193	133
	Calcareous	373	518	196	22
	Organic				
High altitude	Siliceous	506	398	117	9
	Mixed	95	103	120	88
	Calcareous	248	159	40	10
	Organic				

Table 3.1: Typology of water bodies in France; most common ones (80%) are highlighted in green. Source: Nixon et al, 2012

For river water bodies and per measurement point or centroid (the analysis method with the most data available), most bodies occur as small high-altitude, and medium-sized mid- and low-altitude types, implying a differentiated network without much convergence. Notable is a general drop in the occurrence of all types from mid- to low-altitude; this discrepancy can be due to the very large category still having high differentiation with very large lowland rivers having much higher discharge, or heavy evaporation, extractions or groundwater recharge.

Among geology, siliceous is by far the most occurring type, with the mixed type hardly present and only a single organic type identified – a small lowland river water body. This might imply that diversification is highest at medium altitude; at lower ground the rivers converge and river bodies at higher altitude are either not common or too small to be counted. Notable is the lack of organic river bodies – only one has been identified, a small lowland river body.

4. Ecological and chemical status

Ecological status

The WFD states that member states have to assess the biological, physico-chemical and hydromorphological status of river water bodies, ranging from high ('pristine') to good, moderate, poor and bad, culminating in an ecological status classification determined by the lowest scoring sub status. For the biological status assessment, France has developed methods for only four biological quality elements. A fifth, macrophytes, is partially developed. These methods are considered by France to be representative, although the number is low and 4 of these apply to river water bodies. The developed BQE's are shown in table 4.1 below:

RBD	Rivers							Lakes							Transitional							Coastal						
	Phytoplankton	Macrophytes	Phytobenthos	Benthic invertebrates	Fish	Physico-Chemical	Hydromorphological	Phytoplankton	Macrophytes	Phytobenthos	Benthic invertebrates	Fish	Physico-Chemical	Hydromorphological	Phytoplankton	Macroalgae	Angiosperms	Benthic invertebrates	Fish	Physico-Chemical	Hydromorphological	Phytoplankton	Macroalgae	Angiosperms	Benthic invertebrates	Physico-Chemical	Hydromorphological	
FRH															*	*	*	*	*			*	*	*	*			
<i>* not normalised protocol has been developed</i>																												
<i>Assessment methods fully developed for all BQEs</i>																												
<i>Assessment methods partially developed or under development for all or some BQEs</i>																												
<i>Assessment methods not developed for BQEs, no information provided on the assessment methods, unclear information provided</i>																												
<i>- Water category not relevant</i>																												

Table 4.1: Availability of biological assessment methods. Source: RBMP's,

Assessment of physico-chemical and hydromorphological characteristics have only been partially developed. French RBMP's do state the level of uncertainty on the classification of ecological status, based on data availability and coherence with stated pressures, and an attempt to increase the level of certainty of classifications is ongoing. The reported ecological status of surface water and artificial water bodies for the whole of France is shown in table 4.2 below.

RBD	Total	High		Good		Moderate		Poor		Bad		Unknown	
		No.	(%)	No.	(%)	No.	(%)	No.	(%)	No.	(%)	No.	(%)
natural surface water bodies	1630	53	3.3	430	26.4	764	46.9	279	17.1	91	5.6	13	0.8
artificial and heavily modified water bodies	120	1	0.8	11	9.2	22	18.3	35	29.2	13	10.8	38	31.7

Table 4.2: Ecological status and potential of surface water bodies. Source: WISE

Closer analyses of biological, physio-chemical and hydrogeographical quality status for the Seine-Normandy basin have been published and of these one will be presented here. Figure 4.1 on the next page shows biological quality as indicated by fish populations.



Figure 4.1: Biological quality of watercourses, Fish Index 2001. Source: DIREN, AESN

Generally, fish populations degrade towards the center of the basin. This can be explained by the concentration of industrial activity and population density around the central river channels. A similar pattern, although more complex due to different sensitivities, can be observed for macroinvertebrates, hydromorphological elements, organic pollutants, 'métox' (wastewater and industry) pressures and metal pollutants. Pesticide contaminations are spread out more evenly over the entire basin, since agriculture isn't focused in the urbanised areas.

Chemical status

As per the WFD, chemical status also has to be assessed, resulting in a 'good' or 'fail' status based on the concentration of the substance with the highest positive deviation from the reference norm (one-out all-out method).

In the Seine-Normandie basin, the assessment has been performed based on the Annex I set of substances. For the entire basin, data has been published on organic pollutants, metal concentrations, pesticides, micropollutants and radioelements. However different subsets of the full Annex I set have been used per region and it is unclear which substances have been used for the assessment of water body chemical status. It is also unclear whether chemical status has been determined for sediment and biota, whether background concentrations have been taken into account, and if the bioavailability factors of metals have been included. Certain projects have been started to systematically collect data on pollutants, yet the main focus in reports is on measures and legislature limiting emissions, not on quantifying the concentrations currently present.

RBD	Total	Good		Poor		Unknown	
		No.	%	No.	%	No.	%
<i>natural surface water bodies</i>	1630	98	6.0	1120	68.7	412	25.3
<i>artificial and heavily modified water bodies</i>	120	5	4.2	44	36.7	71	59.2

Table 4.3: Chemical status of surface water bodies. Source: WISE

Table 4.3 above and figure 4.2 below show overall chemical status; note that the table contains data for all of France.

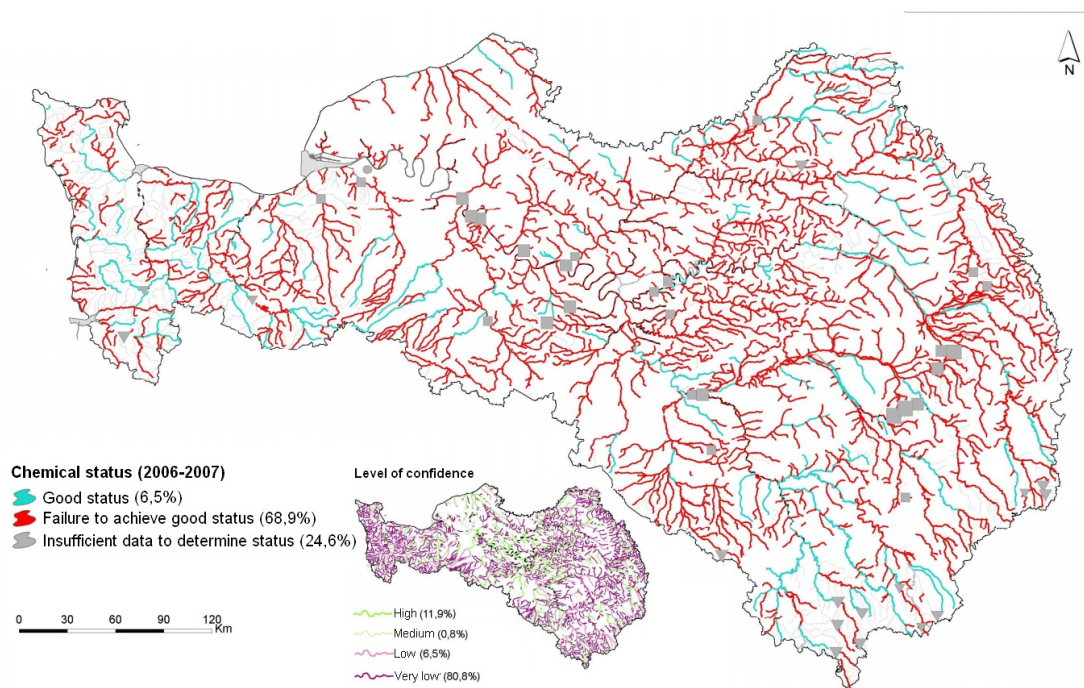


Figure 4.2: Chemical status for the Seine-Normandie Basin. Source: SDAGE 2010, Annex 4

It is notable that PAH's and the compound DEHP occur very often; when disregarded in the extrapolation the percentage of river water bodies that classify as 'good' rises with 7,8%. Whether these are the main goal of measurements will be discussed later.

River water body, case study

One river water body will be discussed in detail. Figure 4.3 below shows the location and general topography of the segment.



Figure 4.3: General location and topography of the river water body case study, FRHR230B

This river water body is designated FRHR230B, it comprises a segment of the river Seine from the merger with the Mauldre to the merger with the Epte (neither merging points are included) and it is situated about 40 km's downstream of Paris. The main source of data for this analysis is from the WFD surface water viewer of the EEA. It is a heavily modified river water body, mainly due to channelization, with a length of 42172.9 meters. The segment has been chosen to illustrate the effect of dense urbanized area and because data is available for the segment. Although certain reports contradict each other, the most detailed have been used. The segment is classified as having overall 'bad' ecological potential. The river segment upstream closer to Paris has a higher ecological classification.

As for quality elements, table 4.4 below shows detailed classifications and the associated confidence level.

Ecological status or potential:		potential (Artificial or heavily modified water bodies)						
Ecological value:								
Phytoplankton	Other Aquatic Flora	Macroalgae	Angiosperms	Macrophytes	Phytobenthos	Macroinvertebrates	Fish	Other Biological QE
Hydromorph	General Physico Chemi	Non Priority Specific Pollutants	Other National Pollutants					

High
 Good
 Moderate
 Poor
 Bad

Unclassified
 Not applicable
 No data

Status confidence: medium confidence

Table 4.4: Ecological potential for FRHR230B. Source: WISE

The combined classifications would result in an ecological potential of 'poor', not 'bad'; the ecological value apparently refers to overall ecological potential including chemical status, to be discussed later. Exemptions have been granted for phytoplankton and other aquatic flora, benthic invertebrates and fish, physico-chemical and hydromorphological elements based on technical feasibility, disproportionate costs and natural conditions not allowing for improvement in time.

The segment also fails to achieve good chemical status. Heavy metal, pesticide and industrial pollutant concentrations are all within maximum limits; the failure is noted by 'other pollutants'. A more detailed report of the sector ("Liste complète des mesures sur la direction de secteur", 2007) states the substances causing failure of ecological status:

- benzo(g,h,i)perylene
- indeno(1,2,3-cd)pyrene

These substances have been classified as 'priority hazardous substances' (QESD, 2008). An exemption has been given for the chemical status as per article 4.4, when natural conditions don't allow improvement in time.

The combination of 'moderate' ecological potential but 'fail' chemical status has lead to the overall ecological classification of 'bad'.

Several exemptions have been defined as an IPPC, two undefined point-source pollutions, a storm overflow and a manufacturing abstraction point. These probably refer to, amongst others, the power plant at 48°58'N, 1°45'E ("Le Centrale"), the wastewater treatment plant at 49° 0'N, 1°39'E ("OTV Holdings") and the steel factory at 49° 2'N, 1°33'E ("Iton Seine"). Another notable element is the dam-lock at 49° 1'N, 1°37'E ("Barrage-écluse de Méricourt, figure 4.4).

These industries and structures, together with the heavily modified waterway and the intensely urbanized areas upstream, are probably the main causes the river water body is failing its ecological assessment. The stated expectancy for 2015 is nevertheless 'good' overall ecological status.



Figure 4.4: Barrage-écluse de Méricourt. Photo: J.F. Murie

5. Current and future threats

The past 40 years

Back in the early 1960's the condition of the river Seine was very poor. Many species of fish had disappeared, plants were dying and the overall water quality was poor. Since that time the river has restored significantly. This change was mainly caused by the creation and adoption of the new Water Law in 1964 (United Nations, 2003). A good example of the effectiveness of this law is the fish recovery. In the 1960s only three of the 32 endemic species could be seen on a regular basis in Paris. Today the Seine has more than 20 endemic fish species. The improvement in water quality is partly due to the installment of more than 500 new wastewater treatment plants, lowering organic pollution, the loss of nutrients to the river and harmful industrial discharge. However, despite these improvements problems still remain.

Nitrate issues

One major problem is the nitrate situation. The problem with nitrates is one of the few issues that has worsened since the 1964 Water Law. Nitrates can cause eutrophication and the proliferation of toxic micro-algae. This in turn effects the water quality which can have negative effects on the biology of the river and humans. Eutrophication often happens when phosphate levels are also high. Phosphate inputs have decreased since the 1960s but are still considered too high. Nitrates can also cause pollution in coastal areas. The main source of the nitrates is from agriculture. The Seine-Normandy basin is a very active area when it comes agriculture. The (excessive) use of fertilizers by the many farms in the area allows nitrates to enter the rivers by surface run off or by groundwater. Figure 5.1 shows the Seine-Normandy basin and the amount of arable land.

Nitrate concentrations in the groundwater are also a concern. Although these concentrations are still below the limit for producing drinking water it may become a serious problem in the future if not actions are taken. Figure 5.2 shows the nitrate contamination in aquifers. As can be seen from this figure the maximum does not exceed 50 mg/L. These concentrations cannot exceed 100 mg/L, which is the threshold for nitrates in drinking water (United Nations, 2003).

Pollution from industry

In the Seine-Normandy basin, the industry plays a big part alongside agriculture. Roughly 60% of the automobile industry and 37% of the oil refining industry of France is located in the basin. The basin is responsible for 40% of the national industrial production. Fortunately, the pollution that was caused by the

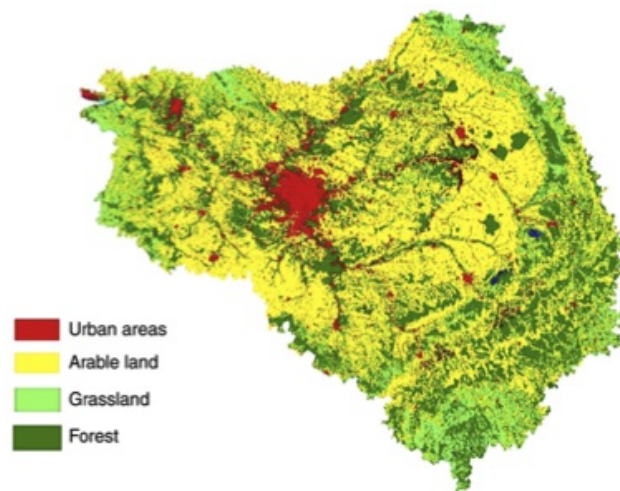


Figure 5.1: Amount of arable land in the Sein-Normandy basin. Source: Billen et al, 2007

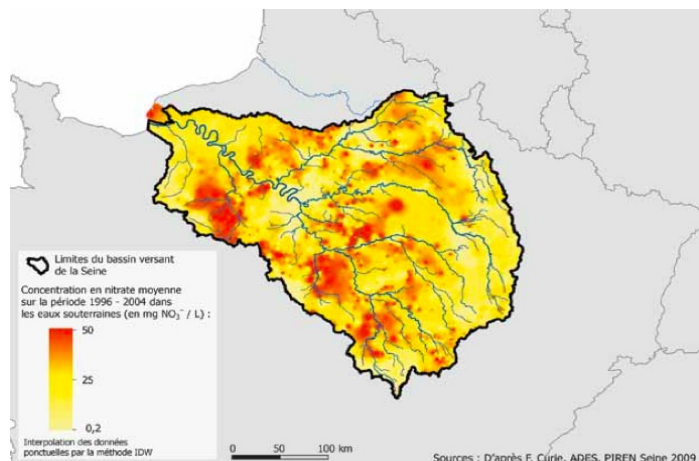


Figure 5.2: Nitrate concentration in aquifers of the Seine-Normandy basin. Source: F. Curie, ADES, PIREN, 2009

industry has also improved since the 60's despite the large amounts of hazardous waste that is created. The reason for this is the improvement and increase of waste treatment plants and disposal facilities, which are very effective in treating hazardous waste. However, accidents are inevitable. And every year about 30 incidents a year happen in the industry that cause contamination (United Nations, 2003). So even though the industry has managed to improve treatment of industrial discharge, it is still considered a current and future threat.

Other pollutants

There are two other chemicals that are a threat to the water quality of the Seine. One is polychlorinated biphenyl or PCB. PCB are industrial product that can be produced when domestic waste is incinerated, or with the production of paints. PCB's are considered toxic to both animals and humans. Despite decreasing, PCB concentrations are six times higher than the national average (United Nations, 2003). Another toxic substance that is of concern is triazine. Triazines are used in pesticides which is used mostly by agriculture.

Pollution due to flooding sewers

Another issue is the possibility of urban runoff pollution. During storms, urban runoff can be the cause of contamination of the river Seine. Also, when extreme rain events occur they can flood the sewer system. The result is that waste water ends up in the Seine. The combination of already water already containing urban surface pollution and waste water from sewer system can cause significant contamination of the river (Even & Poulin et al., 2004). The flooding of the sewer system can happen when a city makes use of a combined sewer. A combined sewer makes use of a single pipe system for waste water and stormwater runoff. Unfortunately the city Paris uses just such a system. Therefore the risk of the sewers overflowing is still a threat.

Climate change

Then there are the effects of climate change. Research has been done by Ducharme et al. to investigate the possible effects that climate change might have on the Seine-Normandy basin. The conclusion of this research is that the Seine river system could suffer more from changes in for instance demography or land use in the coming 50 years than would the effects of climate change. Because of the intensity of agriculture and industry, the threats from those two combined is more significant than are the possible effects of climate change. Changes in especially agriculture will need to change if conditions are to improve over the next 50 years.

6. Administrative organization of Seine River and costs

Water use in France is divided into three sectors: State, territorial and community users., which Overlaps and intertwines with waters governance. At the national level the National Water Committee, consulted on policy and on the drafts laws and regulations, is governed by members of parliament and by some elected representatives of socio-economic groups and other associations (Richard & Bouleau et al., 2010). The French National Water Committee unites users, councilors, associations, catchment committee chairpersons and the State. It is the body that defines a preliminary direction for defining the public water policy at national level.

Basin level

At the basin level the Seine-Normandy basin is governed by a basin committee and water agency, which is organized into three locally elected territorial levels: municipalities, counties and regions. These groups are comprised of elected members of the socio-professional associations and the State committee. The Seine-Normandy basin includes 185 members of which, 40% represent the community, 40% water users and 20% represent the state (Seine-normandie.eaufrance.fr, 2014). Management per catchment area is reflected in water agencies and in local water commissions bringing together local councilors as well as business', associations and the State. (Ministry of Ecology, Sustainable Development, Transport and Housing, 2012). The Water Agency Seine-Normandy is organized into geographical areas to be closer to the water users, therefore being more tailored to there needs. This is divided into 8 regions: Directorate of Normandy Rivers, Directorate of Seine-Aval, Directorate of The Oise Valleys, Directorate of Paris Small crown, Directorate of RIVERS OF ILE-DE-FRANCE, Directorate of the valleys of Marne, Directorate of Seine upstream (Eau-seine-normandie.fr, 2014)

Sub-basin level

At the sub-basins level there are local water commissions, which participates in the development of planning and water management. This consists of a majority of elected officials, water users and administrations. At the local level Mayors are responsible for drinking water supply and sanitation of their common wastewater and storm water. (Seine-normandie.eaufrance.fr, 2014).

Figure 6.1 below provides a general structure to the complex system of Governance of the Seine-Normandy Basin.

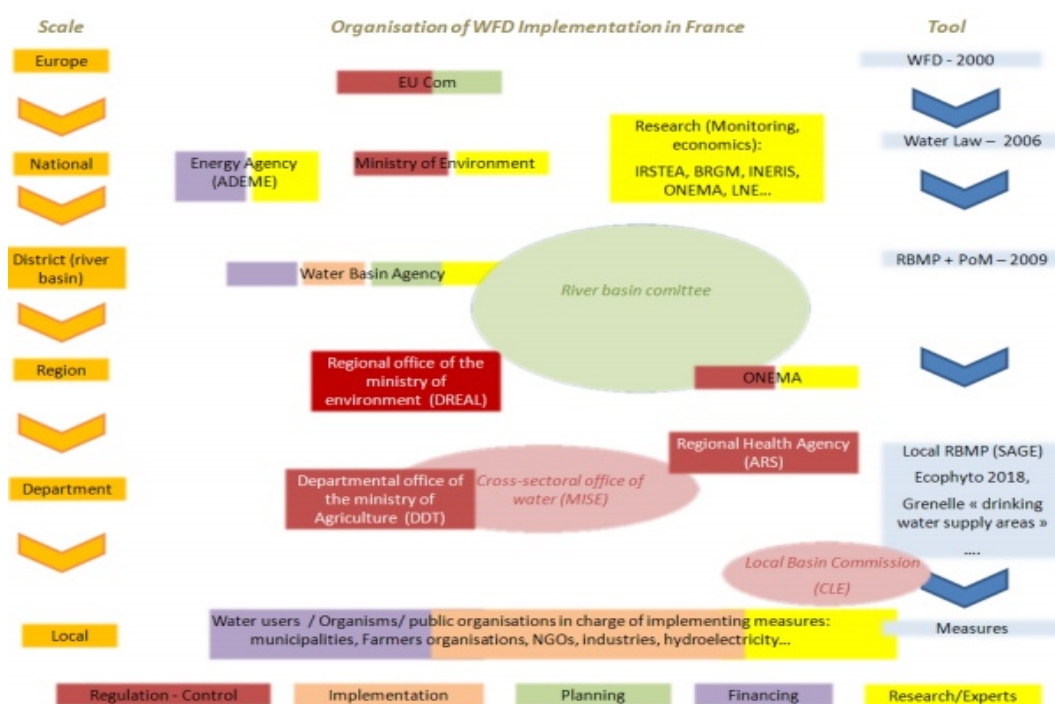


Figure 6.1: Organisation structure of the WFD implementation authorities in France. Source: Water Policy in France, 2012

River Basin Management Plan

The French water management system places great importance on a high degree of local responsibility, public-private sector partnerships, coordination of users on a river basin level and a taking into account of all water uses. Municipalities have been responsible for all services associated with water since the nineteenth century. Today, they are not only responsible for the creation of water services and operating facilities, but are also legally obligated to maintain the quality of services, as well as determining the rates charged to users in their community. Towns often set up inter community associations to operate drinking water supply and wastewater treatment networks. They also create joint public-private partnerships by subcontracting water supply and treatment services to private companies under various types of contracts. All water users must comply with standards set by the water laws, and local representatives of state agencies monitor compliance. The state, therefore, remains the 'guardian' of the resource. (Ministry of Ecology, Sustainable Development, Transport and Housing, 2012)

The Water Agency Seine-Normandy

Faced with this complex allocation of responsibility, the role of the Water Agency at the river basin level is to promote measures undertaken to ensure a balance between water resources and needs. It performs its role via financial vehicles. It allocates funds – in the form of loans or subsidies – for projects that correspond to the objectives of the Water Agency policies, which form the fabric of the water management plan (Eau-seine-normandie.fr, 2014).

The Water Agency Seine-Normandy set targets (SAGE's) for the use, development and protection of water for certain sub sets of the basin. These must be always related the master development and water management scheme (SAGE).

1. Reduce point source pollution of the environment by conventional pollutants
2. Reduce diffuse pollution of aquatic environments
3. Reduce pollution of the aquatic environment by dangerous substances
4. Reduce microbiological pollution environments
5. Protect water catchments to supply current and future drinking water
6. Ensure the public drinking water supply
7. Protect and restore water resources
8. Reduce and prevent the risk of flooding (Eau-seine-normandie.fr, 2014)

Essentially these are the commandments of the agency and embody the detailed water management plan.

An example of the micro level of the water management plan is the SAGE for Aisne Vesle Suiippe which is a management plan tailored to the Niche's most pressing issues.

- Quantitative resource management during low water
- Improving the quality of groundwater
- Improving the quality of surface waters
- Preservation and safeguarding of drinking water
- Preservation and restoration of the quality of aquatic and wetland
- Flooding and runoff
- Management of hydraulic structures (Eau-seine-normandie.fr, 2014)

River basin costs

Water supply and sanitation have a cost. Water needs to be transported, cleaned at treatment plants, re-distributed. Additionally water must be tested to guarantee quality. Users bear the main expenditure of towns and their municipalities in terms of investment in and operation of the necessary equipment, on their water bill. The Seine-Normandie water agencies' budget is based on application of the "polluter pays" principle. The taxes collected are directly proportional to the amount of water used and the level of pollution discharged within the region. Revenue is redistributed by the Water Agency in as interest-free loans or subsidies. These are handed out to local decision makers and business owners: towns or municipalities, industrialists and farmers, in order to finance programmers to fight pollution, manage water resources and reconstitute quality aquatic environments (Webworld.unesco.org, 2014).

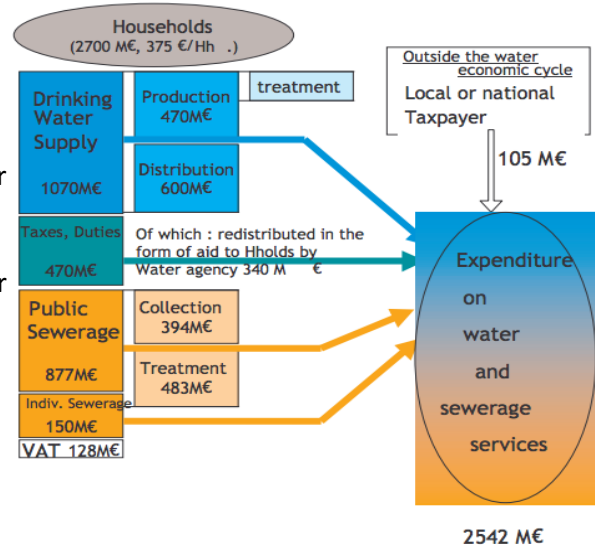


Figure 6.2: Costs of water management. Source: WISE

7. Conclusion

In the above chapters we have seen an outline of the different aspects of a river basin. It has become clear that the task of collecting and organizing data on all the relevant issues pertaining to the WFD is a challenge not only for the member states, but for researchers as well. Clearly this is an area with much room for development, and in the light of the coming difficulties in water management because of climate change and population growth it is hopeful that the task is underway. By framing the transboundary issues involved with water, we can not only provide a safe future but perhaps also lead the way and provide an example for other developing regions to follow. The WFD is a step towards a more better future in many ways.

8. Sources

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